12.1 Functions of two variables

Before calculus III, functions are

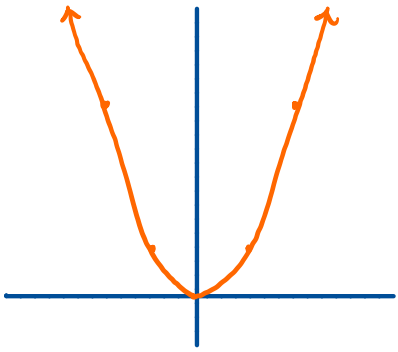
$$\underbrace{y}_{\text{output}} = \overbrace{f}^{\text{name of function}}(\underbrace{x}_{\text{input}})$$

We represented functions in many ways. E.g., if $f(x) = x^2$:

Table

x	y
-2	4
-1	1
0	0
1	1
2	4
$\vdots$	$\vdots$

Graphically

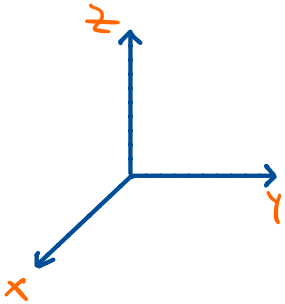


In this class we move to functions of multiple variables. E.g., functions of two variables:

$$\underbrace{z}_{\text{output}} = \overbrace{f}^{\text{name}}(\underbrace{x, y}_{\text{input}})$$

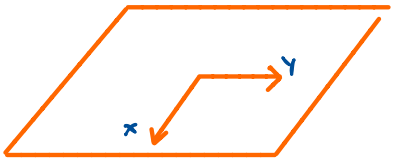
→ Do problems 1-3

How might we represent $z = f(x,y)$ graphically? In here:

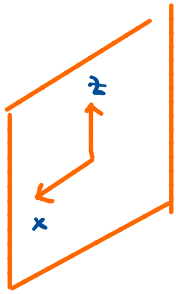


xyz-space, a.k.a. $\mathbb{R}^3$

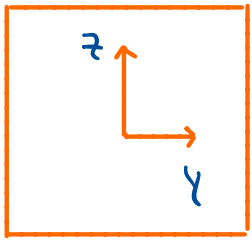
We use the three coordinate planes as landmarks in $\mathbb{R}^3$:



xy-plane



xz-plane



yz-plane

→ Do problem 4.